Recording and analysis of spike trains

Poisson, Fano, CV, STA, & PSTH

Computational Neuroscience

University of Bristol

Based in part on C. O'Donnel's slides

mrule

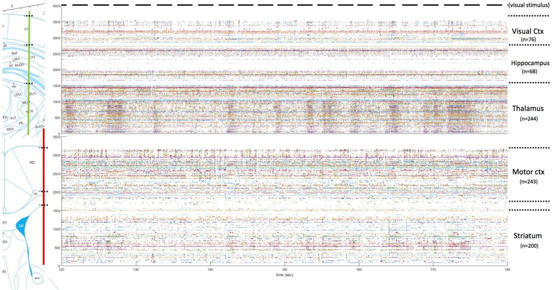
Learning outcomes:

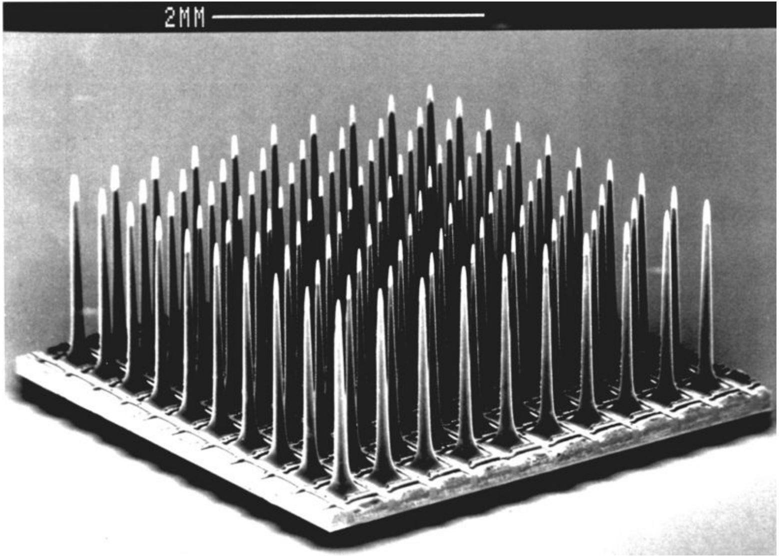
- Extracellular recording
- Understand the basic analyses done on spike trains.
- Be able to explain the mathematics underlying CV, Fano Factor, Spike-triggered average.
- Inhomogeneous Poisson process as a statistical model of a spike train
- Spike train statistics (Inter-spike interval, Fano factor, coefficient of variation)
- Be ready to write computer code that explores the statistical properties of a neuron's spiking output, and how this spiking output relates to external stimuli.

Intracranial electrophysiology

Intracranial electrophysiology

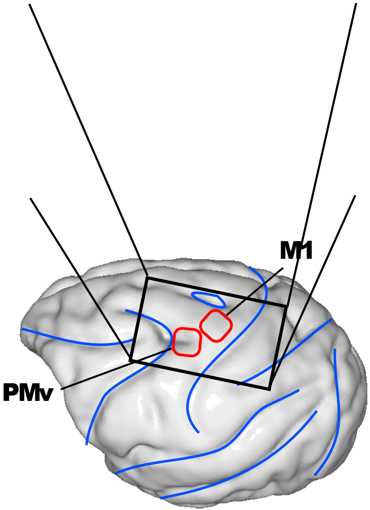
- Electrodes inserted directly into the brain to record field potentials/voltages "extracellularly".
- Invasive, so only used in humans in medical cases. But routinely used in rodents and some non-human primates.
- Excellent temporal resolution (<1ms) and good spatial resolution: with care can resolve single neurons.
- Data in the form of voltage time series but usually processed to extract spike times.
- Neuropixels probes thousands of neurons can be recorded in rodents.

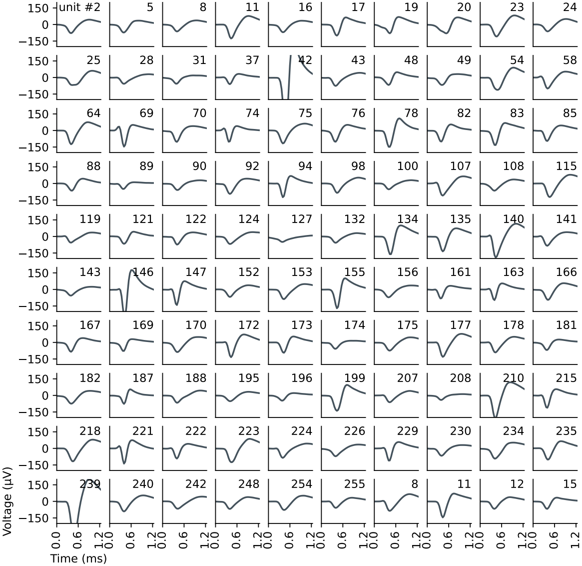




Multi-electrode "Utah" array

(Vargas-Irwin et al. 2010)



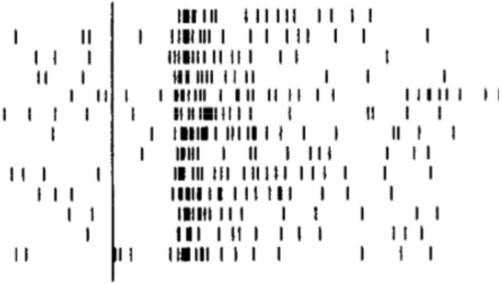


Data from Brochier et al. (2018)

Spike trains

Below is a "raster plot" from a single neuron recording in monkey visual cortex.

- Each tick is a spike.
- Each row plots the neuron's spike train vs time for a differential repeated 'trial'.
- The visual stimulus is identical in each case but the neuron's response has a high degree of trial-to-trial variability.
- The inter-spike interval and the spike count are two measures of neural response.



Motter, J Neurophysiol, 1993

Analysing spiking data

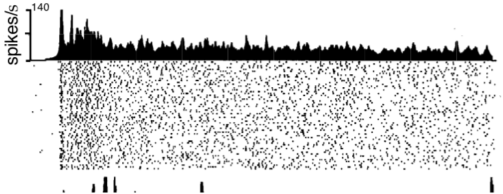
Peri-stimulus time histogram (PSTH)

Neural responses to stimuli are variable, meaning they are not always identical from trial to trial (repeated presentations of the same stimulus).

The **peri-stimulus time histogram** is one way to represent the average spike response across trials.

The idea is:

- Superimpose the neuron's spike responses from multiple trials.
- Bin time into small intervals (e.g. 2 ms).
- Histogram the spike counts in each time bin.



Dayan and Abbott, 2001

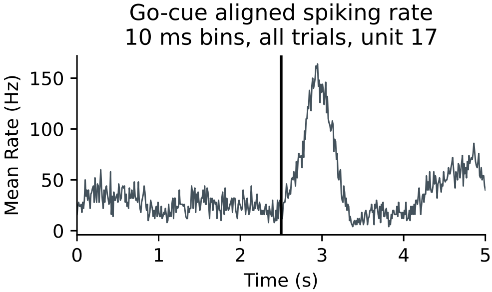
Peri-stimulus time histogram (PSTH)

Record

- Stimulus or other signal (e.g. movement) $x(t)$
- Spike train from a neuron with spike times $t_s \in \{t_1, t_2, \ldots, \}$

Peristimulus Time Histogram (PSTH)

- Align spikes to stimulus onset
- Combine trials
- Take histogram



Spike-triggered average

Another way of analysing the data is to ask the question: what aspect of the stimulus caused the neuron to spike?

One way of quantifying this is the **spike-triggered average** stimulus:

$$S(\tau) = \frac{1}{N}\sum_{i=1}^{N} s(t_i - \tau)$$

where $s(t)$ is the stimulus value, t_i's are spike times and τ is a time interval.

Spike-triggered average

Record

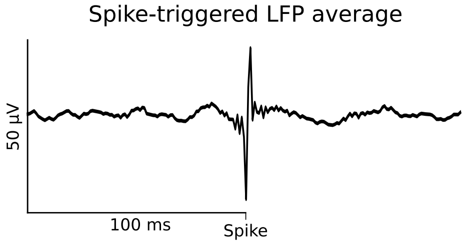
- Stimulus or other signal (e.g. movement) $x(t)$
- Spike train from a neuron with spike times $t_s \in \{t_1, t_2, \ldots, \}$

Spike Triggered Average (STA)

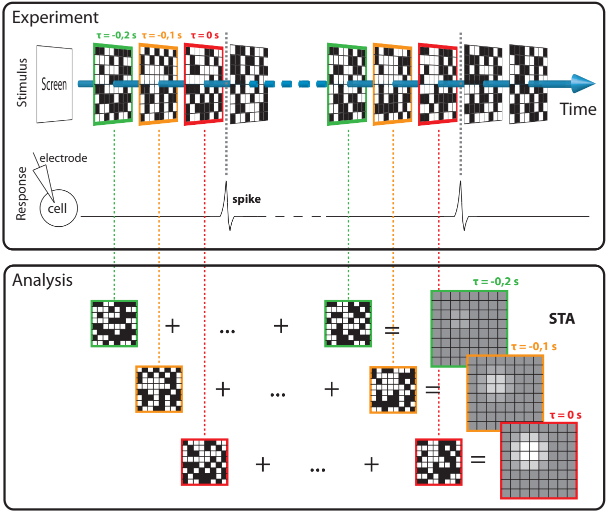
- For each spike time time t_s, get $x(t)$ surrounding spike time $x(t_s + \tau)$, $\tau \in [-\Delta, \Delta]$
- Average these together

Event-triggered average

- Align & average continuous signal



Spike-triggered average (STA)



via wikimedia

Statistial modelling using Poisson processes

How should we model the statistical regularities in spike trains?

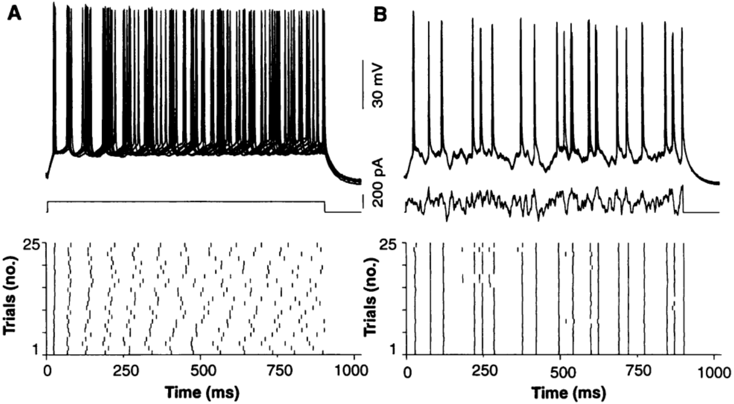


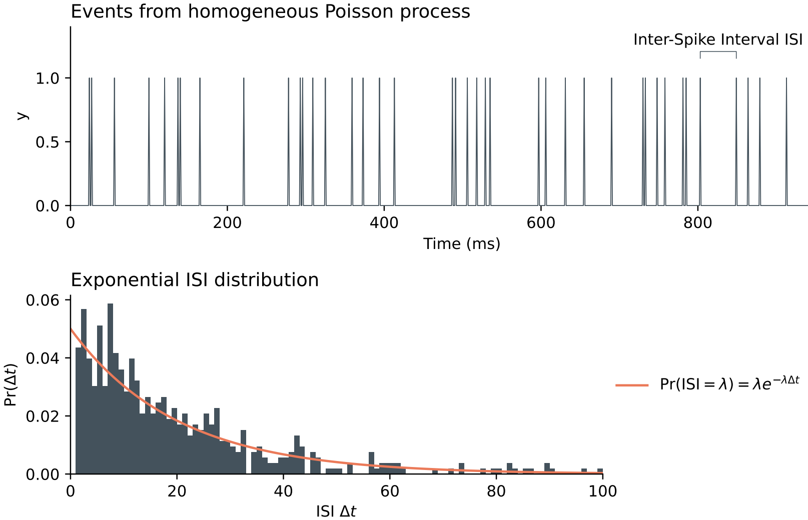
Fig. 1. Reliability of firing patterns of cortical neurons evoked by constant and fluctuating current. (**A**) In this example, a superthreshold dc current pulse (150 pA, 900 ms; middle) evoked trains of action potentials (approximately 14 Hz) in a regular-firing layer-5 neuron. Responses are shown superimposed (first 10 trials, top) and as a raster plot of spike times over spike times (25 consecutive trials, bottom). (**B**) The same cell as in (A) was again stimulated repeatedly, but this time with a fluctuating stimulus [Gaussian white noise, μ_s = 150 pA, σ_s = 100 pA, τ_s = 3 ms; see (*14*)].

Mainen and Sejnowski 1995

Spike Train

- Electrical recordings at e.g. 30 KHz detect action potentials
- Find the action potentials in the voltage trace (spike sorting)
- Count action potentials in e.g. $\Delta t = 1$ ms bins
- 1 KHz binary or non-negative integer timeseries

Homogeneous Poisson process



Homogeneous Poisson process

Spike counts in a given bin follow a Poisson distribution:

$$\Pr(y = k) = \frac{1}{k!}\lambda^k e^{-k}$$

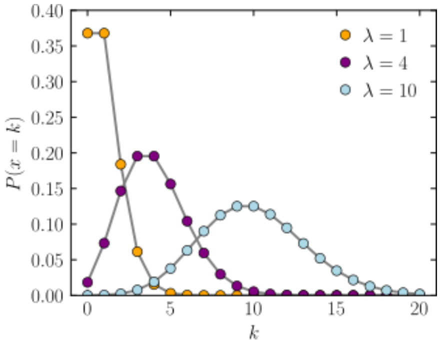
(In the homogeneous case) Inter-Spike Intervals (ISI) follow an exponential distribution

$$\Pr(t_{\mathsf{ISI}} = t) = \lambda \exp(-\lambda t)$$

Poisson process

$$k_t = \# \text{ of spikes in } [t, t + \Delta t) \in \mathbb{Z}_{\geq 0}$$

$$k_t \sim \text{Poisson}(\lambda \cdot \Delta t)$$



Wikimedia Commons

The fano factor and coefficient of variation provide quick spot-checks on the statistical structure of spike trains. Purely random firing will typically look like an inhomogeneous Poisson process, and both of these quantities will be 1. Numbers above 1 typically mean there are additional slow processes affecting the neuron's firing rate. Numbers below 1 can appear with neurons that spike rhythmically or regularly.

Fano Factor

The **Fano Factor** is a statistical measure of the dispersion of a probability distribution.

In neuroscience it is typically used to quantify the variability of spike trains.

It the variance of the spike count divided by the mean spike count:

$$F = \frac{\sigma^2}{\mu}$$

Importantly it usually applied to the spike counts over some time interval (not the interspike intervals).

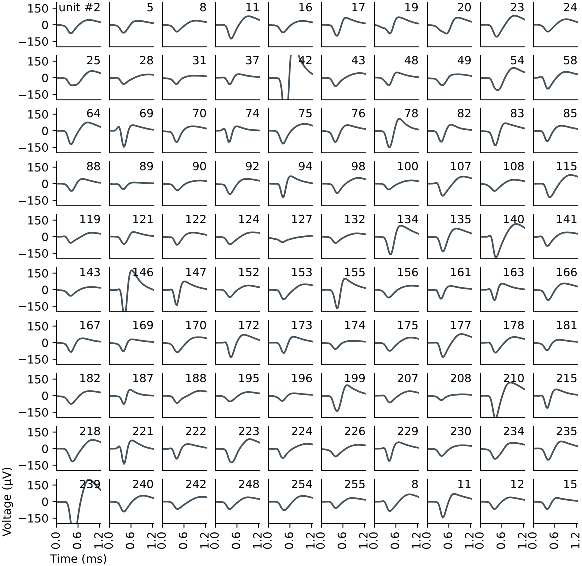
Coefficient of variation (CV)

A different measure of variability is the **coefficient of variation** (CV).

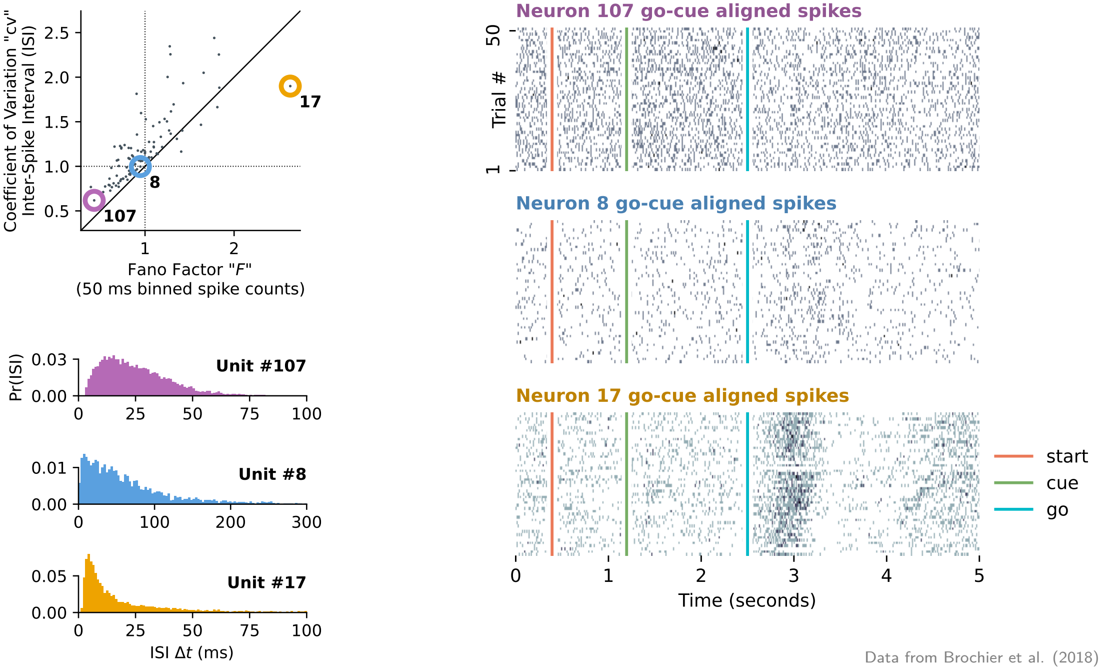
It is the standard deviation of the interspike interval divided by the mean interspike interval:

$$CV = \frac{\sigma_{ISI}}{\mu_{ISI}}$$

In contrast to the Fano Factor, it is usually applied to the interspike intervals (not the spike counts).



Data from Brochier et al. (2018)


Coefficient of Variation "cv"
Inter-Spike Interval (ISI)
2.5
2.0
1.5
1.0
0.5
17
8
107
1
2
Fano Factor "F"
(50 ms binned spike counts)
Pr(ISI)
0.03
0.00
Unit #107
0
25
50
75
100
0.01
0.00
Unit #8
0
100
200
300
0.05
0.00
Unit #17
0
25
50
75
100
ISI Δt (ms)
Neuron 107 go-cue aligned spikes
50
Trial #
1
Neuron 8 go-cue aligned spikes
Neuron 17 go-cue aligned spikes
0
1
2
3
4
5
Time (seconds)
start
cue
go
Data from Brochier et al. (2018)

Homogeneous vs. inomogeneous Poisson processes

Poisson process: $\Pr(\text{spike at time } t)$ depends only on intensity $\lambda(t)$.

- **Homogeneous:** $\lambda(t)$ constant
- **Inhomogeneous:** $\lambda(t)$ changes

Fano factor $F = \dfrac{\sigma_k^2}{\mu_k}$

Coefficient of variation $\mathrm{cv} = \dfrac{\sigma_{\mathrm{ISI}}}{\mu_{\mathrm{ISI}}}$

For homogeneous, $F = \text{cv} = 1$

- Less random? $\text{cv} < 1$ $F < 1$
- More random? $\text{cv} > 1$ $F > 1$

Ask me about Bernoulli processes and "point processes"

Inhomogeneous Poisson

"Real" neurons?

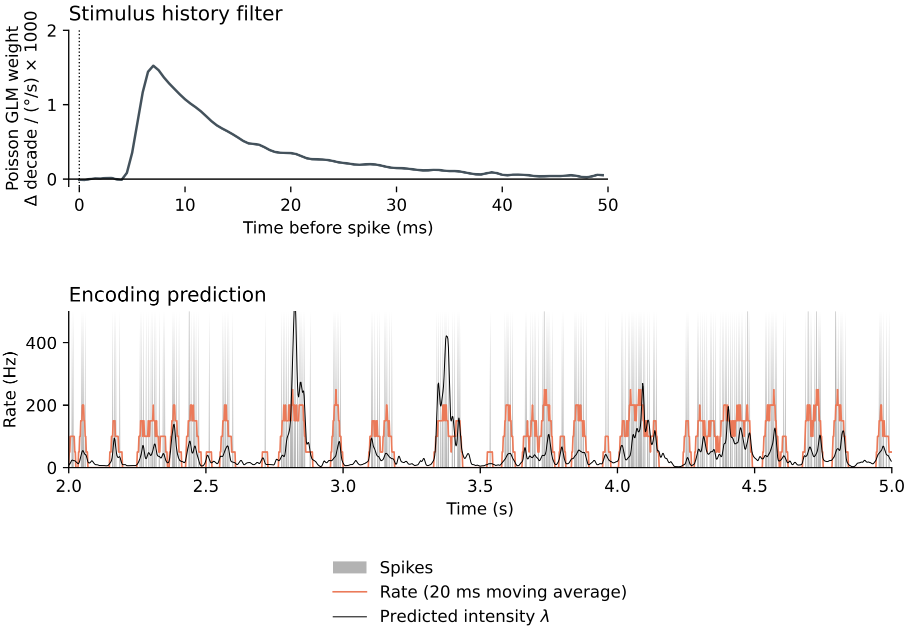
- Refractory $\rightarrow$ less random than the homogeneous Poisson
- Varying inputs $\rightarrow$ more random than the homogeneous Poisson

Inhomogeneous Poisson process

Model as Poisson with rate varying in time $\lambda(t)$

Assume $\lambda(t)$ changes slowly relative to timescale Δt

- $\lambda(t) \approx$ constant for $t \in [t_0, t_0 + \Delta t)$
- k_{t_0} = Poisson$(\lambda(t_0) \cdot \Delta t)$



From the Rob de Ruyter van Steveninck fly H1 data used in coursework

Poisson Generalized Linear Model (GLM)

Predict spikes $y(t)$ from intputs $\boldsymbol{x}(t)$

Rather than using a linear regression:

$$y(t) = \boldsymbol{w}^\top \boldsymbol{x}(t) + c,$$

use a Poisson regression (Poisson generalized linear model)

$$\ln \lambda(t) = \boldsymbol{w}^\top \boldsymbol{x}(t) + c$$
$$\int_t^{t+\Delta} y(t)\,\mathrm{d}t \sim \mathrm{Poisson}\left(\int_t^{t+\Delta} \lambda(t)\,\mathrm{d}t \right).$$

Binned spikes counts are *conditionally* Poisson

$$\ln \Pr(y_\Delta) = y_\Delta \ln(\lambda_\Delta) - \lambda_\Delta - \ln(y_\Delta!)$$
$$y_\Delta = \int_t^{t+\Delta} y(t)\,\mathrm{d}t$$
$$\lambda_\Delta = \int_t^{t+\Delta} \lambda(t)\,\mathrm{d}t$$

We can also add the neuron's own past spiking outputs to the model with a history filter $h(t)$

$$\ln \lambda(t) = \boldsymbol{w}^\top \boldsymbol{x}(t) + c + \int_{\tau>0} h(\tau) y(t-\tau)\,\mathrm{d}\tau$$

This is an exercise in the extended coursework

One simple way to approach it is to treat the *past* spiking output at various time lags as if it were a stimulus input

Bayes' Rule

Decoding a stimulus from spikes using Bayes' Rule

Imagine simulus x and spikes y

Conditional probability

$$\Pr(x, y) = \Pr(x|y)\Pr(y) = \Pr(y|x)\Pr(x)$$

Rearrange to get Bayes' rule

$$\Pr(x|y) = \Pr(y|x)\frac{\Pr(x)}{\Pr(y)}$$

We usually work with log-probability for numerical stability

$$\ln\Pr(x|y) = \ln\Pr(y|x) + \ln\Pr(x) - \ln\Pr(y)$$

If we've observed a specific spiking pattern y, it's common to drop the $\ln\Pr(y)$ part

$$\ln\Pr(x|y) = \ln\Pr(y|x) + \ln\Pr(x) + \text{constant}$$

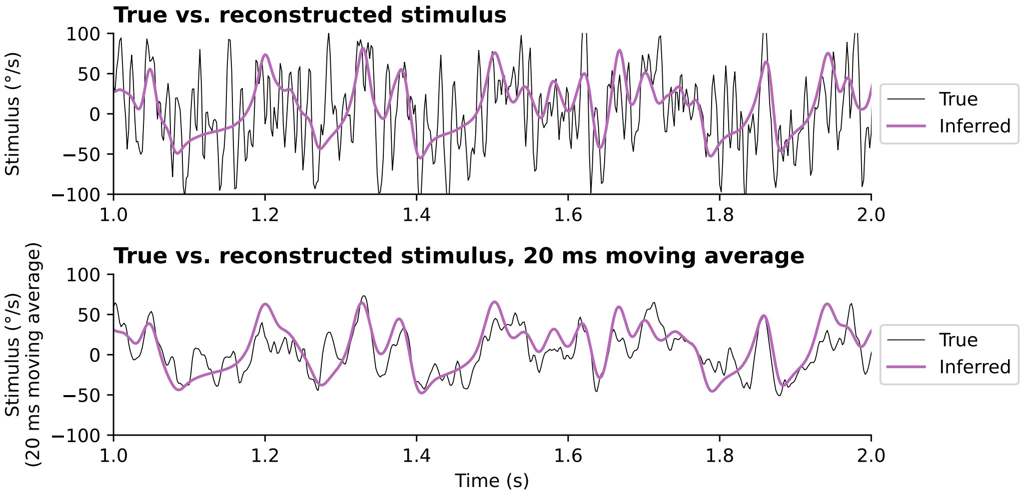
Decoding a stimulus from spikes using Bayes' Rule

Bayes' rule allows us to estimate a stimulus x by combining a prior $\Pr(x)$ with a likelihood $\Pr(y|x)$ of observing a specific pattern of spikes y for each stimulus x.

Rob de Ruyter van Stevenink fly H1 data used in coursework

- You should be able to sample something like the stimulus $x(t)$ by:
  - Drawing i.i.d. standard Gaussian noise $q_i \sim \mathcal{N}(0, 1)$ for each 2 ms time bin
  - Convolving this timeseries with a Gaussian kernel with $\sigma = 4$ ms (i.e. 2 bins)
  - Multiplying by 200
- This gives a multivariate Guassan prior on the signal $\boldsymbol{x}$

This isn't required for the coursework, but you could in principle combine the above prior with the encoding model you will build to make a decoding model, which will let you read off the motion stimulus from the spike train.



Incorporating a spike (auto)history model to capture refractory effecy may improve this

Pause

Recording and analysis of spike trains

____ (t/f): Extracellular recordings are good for measuring membrane voltage and ionic currents

____ (t/f): Extracellular recordings can capture spike trains.

The ______________________________ measures the average number of spikes relative to a time-locked stimulu

The __________________________ measures the average stimulus at various time-lags relative to a neuron's spiking

The ratio σ^2/μ is the __________, and we often apply it to ________________

The ratio σ/μ is the ___________________, and we often apply it to ______________________

The time between spikes in a *homogeneous* Poisson process is ___________ distributed

A Poisson generalized linear point-process model predicts the ________ of a neuron's spiking intensity as a linear function of covariates.

Recording and analysis of spike trains

False (t/f): Extracellular recordings are good for measuring membrane voltage and ionic currents

True (t/f): Extracellular recordings can capture spike trains.

The peristimulus time histogram, PSTH measures the average number of spikes relative to a time-locked stimulu

The Spike-Triggered Average (STA) measures the average stimulus at various time-lags relative to a neuron's spiking

The ratio σ^2/μ is the Fano factor, and we often apply it to binned spike counts

The ratio σ/μ is the coefficient of variation, and we often apply it to Inter-Spike Intervals (ISIs)

The time between spikes in a *homogeneous* Poisson process is exponentially distributed

A Poisson generalized linear point-process model predicts the logarithm of a neuron's spiking intensity as a linear function of covariates.

Aside: Bernoulli Process

Since neurons have a maximum firing rate, we can choose Δt small enough so that all time bins contain either 0 or 1 spikes. Binary spike trains can be modelled as coin flips (Bernoulli) process. For small Δt the Poisson and Bernoulli models behave similarly. The math for the Poisson process is slightly simpler, but the maximum rate limit of the Bernoulli process can be useful. Bernoulli–process models can be fit as logistic regression, where the probability p is a linear-nonlinear function $p = f(\boldsymbol{w}^\top \boldsymbol{x} - \vartheta)$ of regression features $\boldsymbol{x}$, and $f(\cdot)$ is the sigmoidal logistic nonlinearity $f(a) = [1 + \exp(-a)]^{-1}$. You may also see binary models fit using "probit" regression, which is similar to logistic regression but uses the sigmoidal nonlinearity taken from the cumulative distribution function of a standard normal distribution. There are certain applications where logistic vs. probit regression are more convenient. Probit regression of binary spike trains is related to a class of models called the "dichotomized Gaussian", which can be used to model correlations in spiking population activity.

end

page left blank for notes

page left blank for notes